



Quantum Communication Controller

1. Product Overview

The Quantum Communication Controller is a compact, indigenous, high-performance FPGA-based development platform designed as a complete control system for next-generation quantum communication. It features a high-performance SoC integrating multi-core ARM processors and a real-time processing unit. It supports multiple QKD protocols on a single scalable hardware platform offering low-jitter, flexibility, and stable performance required over long-distance fiber links.

2. Technical Summary

Processing System

- Quad-core ARM Cortex-A53 APU
- Dual-core ARM Cortex-R5 RPU
- ARM Mali-400 MP2 GPU

Programmable Logic

- 1.143K Logic Cells, 1,045K Flip-Flops
- 70.6 Mb Block RAM, 1968 DSP slices
- AXI-based PS-PL high-bandwidth interconnects

Memory

- 4 GB PS DDR4 (64-bit SODIMM)
- 512 MB PL DDR4 (x16 devices)
- 8 KB I2C EEPROM
- Dual 64 MB QSPI Flash
- Micro-SD support

Connectivity

- 2x Gigabit Ethernet
- 4x SFP+ cages
- USB 3.0, USB 2.0 OTG
- CAN, SATA, UART-USB bridge
- PMOD, DIP switches, LEDs, reset buttons

Multimedia

- HDMI 2.0
- DisplayPort



High-Speed Transceivers

- Supports PCIe, SATA and high-speed serial links

QKD Interfaces

- NIM & LVTTTL SPD interfaces with the time-to-digital converter
- Dual TDC7200 on board
- Clock data recovery for time synchronization
- Drive SMA interfaces for modulators, SPD gates, and RF inputs
- GTX-based high-frequency interfaces for modulator drive, etc.
- 14-bit DACs for IM, PM, DLI, and VOA bias control
- Three integrated optical switches
- 12-bit ADC for optical power monitoring

Why QCC?

- Indigenously developed AMD MPSoC based FPGA development board
- Standalone solution as integrated Quantum Communication control platform
- Multiprotocol support demonstrated for Quantum Key Distribution
- Eliminates dependency on external components
- High-speed, precise, modular, and scalable
- Suitable for advanced research and deployment

3. Add on

The controller includes configurable time-synchronization IP cores that enable precise alignment of clocks, triggers, and timestamps. These IP blocks can be adapted to various synchronization architectures and deployment needs, ensuring deterministic and accurate timing performance across the entire system.